

Big Data, Bigger Bags



Project 2 - Data Visualization in Tableau

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Business Scenario, Users, & Weekly Decisions

Business: We are a creator management and brand partnership agency that helps brands strategically collaborate with high-reach YouTube creators, musicians, and celebrity channels to maximize campaign visibility and engagement impact.

Primary Dashboard User: Director of Brand & Creator Partnerships

The dashboard is designed to help our Director of Creator Partnerships make weekly strategic decisions about which external influencers show strong growth and engagement momentum, which creators demonstrate high monetization potential for brand partnerships, and which market segments offer the best expansion opportunities for our agency.

Key Weekly Decisions:

Partnership Selection Decision:

Which celebrity or high-reach creator should we recommend to brands based on reach, engagement efficiency, and audience scale?

Campaign Fit Decision:

Which creator categories (music, entertainment, lifestyle, etc.) provide the strongest visibility-to-engagement ratio for upcoming brand campaigns?

Value Screening Decision:

Which large channels appear overvalued (high subscribers but weak engagement) versus high-efficiency channels that offer better partnership ROI?

Strategic Theme & Question:

Optimize brand partnership strategy by identifying high-impact celebrity and creator channels with strong reach-to-engagement efficiency. **How can we identify high-reach celebrity and creator channels that provide the strongest visibility-to-engagement efficiency for brand partnership opportunities?**

What Success Looks Like + KPIs				
Success means identifying creators who combine scale, engagement quality, and efficiency.				
Engagement Rate (%)	Views per Subscriber	Content Performance per Video	Total Channel Subscribers	Partnership Readiness Index (Composite KPI)
Formulas				
$(\text{Number of Likes} + \text{Number Comments}) \div \text{Video Views}$	$\text{Video Views} \div \text{Total Channel Subscribers}$	$\text{Total Channel Views} \div \text{Number of Videos on the Channel}$	Measures reach scale and campaign visibility potential	$\text{Engagement Rate} \times \text{Views per Subscriber}$
Targets				
Above dataset average	Above dataset median	Top 25%	Within the top tier for brand exposure	Above benchmark
Measures				
Measures audience interaction quality.	Measures audience activation.	Measures long-term content efficiency.	Measures reach scale and campaign visibility potential	Identifies high-impact collaboration candidates.

Data & Metric Integrity

Data Source and Time Period: The Kaggle dataset used is the YouTube Influencer Analysis, based on publicly available YouTube channel statistics. The data represents a cross-sectional snapshot of channel performance rather than a time series, meaning it reflects performance at a single point in time instead of tracking changes over multiple periods.

Key Fields Used										
Video Views	Video Title	Creator Name	Creator Gender	Total Channel Subscribers	Total Channel Views	Duration of Video	Duration in Seconds	Date of Video Upload	No of Likes	Language of the Video
Subtitle	Video Description	Hashtags	No of Comments	Date of the Last Comment	Maximum Quality of the Video	No of Videos the Channel	No of Playlist	Premiered or Not	Community Engagement (Posts per week)	

Known Limitations: The dataset reflects a single snapshot in time and may be outdated, meaning current channel performance could differ from what is reported. Public metrics such as views, subscribers, likes, and comments can also be artificially inflated through bot activity or coordinated engagement tactics, introducing potential inaccuracies. As a result, the data may not fully represent authentic audience behavior or true creator performance, and partnership decisions should not rely solely on these surface-level metrics without additional validation.

KPI Definitions
Engagement Rate percent equals No of Likes plus No of Comments divided by Video Views.
Views per Subscriber equals Video Views divided by Total Channel Subscribers
Average Views per Video equals Total Channel Views divided by No of Videos the Channel.
Total Channel Subscribers measures reach scale and potential campaign visibility.

Creator Management & Brand Partnership Analytics Dashboard

Date Last Updated 2/28/2026 10:12:54 PM

Start date

End date

Date of Video Upload

All

Creator Gender

Gender

All

Language Segment

Language o..

Multiple values

Video Quality

Quality

All

Premiere Status

Status

All

Average Engagement Rate (%)

1.15%

Average Views per Subscriber (ratio)

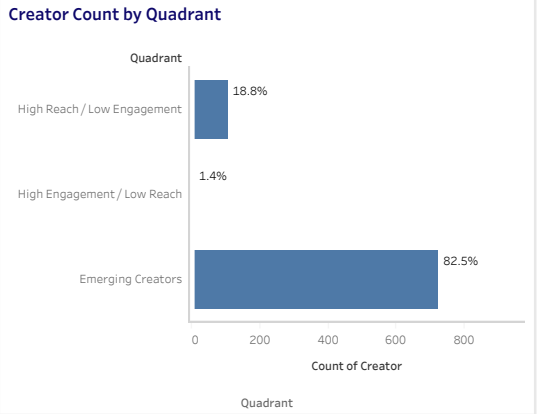
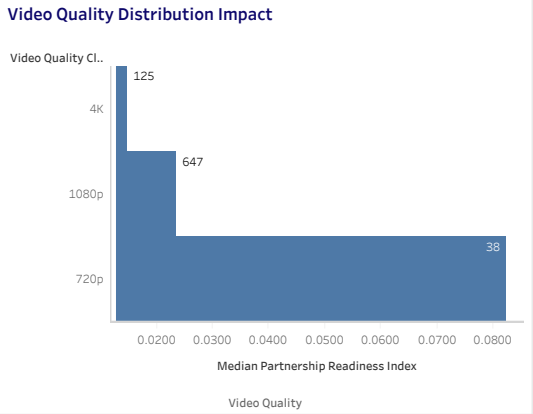
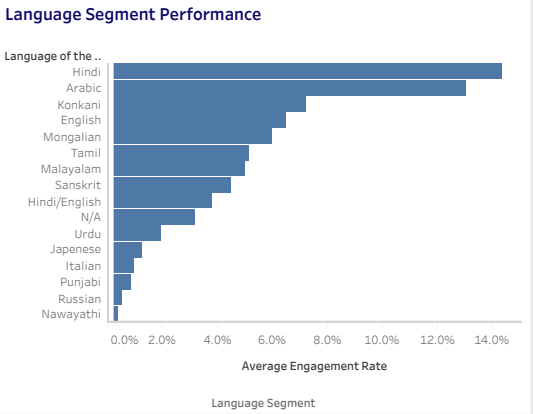
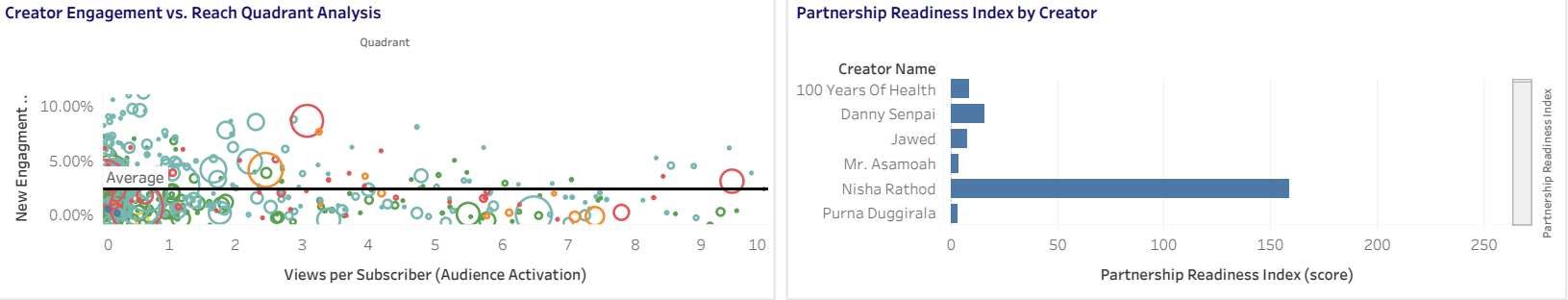
5.212

Average Views per Video (count)

460,558

Top Creator Partnership Readiness Index (score)

0.05975



Creator Partnership Prioritization Table					
Creator Name	Max. Total Channel Subscribers	Avg. Views per Subscriber (Row)	Avg. Engagement Rate (Row)	Avg Views per Video (k)	Avg. Partnership Readiness Index (Row)
UFC - Ultimate Fighting Championship	14,200,000	23.98%	0.75%	449,927	0.0014
The Viral Fever	11,300,000	50.24%	3.92%	3,714,417	0.0159
The Kratos Patti	98,300,000	18.91%	4.79%	3,179,783	0.0090
Thanum Kumar	966,000	93.68%	7.18%	170,088	0.0645
Tanmay Singh	4,510,000	60.64%	12.95%	630,834	0.0619
Sri Ram	7,610,000	103.89%	4.54%	10,386,635	0.0417
Rajalalith	29,200	520.34%	3.44%	41,529	0.1079
Rahul M.	438,000	12.54%	6.75%	90,473	0.0084
Oliver Velez	254,000	12.14%	4.37%	13,317	0.0048
Marvel Entertainment	1,960,000	11.74%	4.10%	643,059	0.0042
Kadhiravan Javachandhiran	524	20.62%	4.76%	42	0.0074
Jay Shetty	4,500,000	22.18%	4.22%	334,131	0.0088
HGOE	133,000	489.28%	4.01%	335,570	0.1444
History	10,700,000	6.33%	1.72%	393,954	0.0008
Hari baskar	2,270,000	111.29%	6.40%	2,275,839	0.0537
Ganesh Prasad	2,250,000	81.06%	3.85%	127,859	0.0284
Forced in Fire	403,000	27.55%	1.80%	395,281	0.0035
Epicurious	4,600,000	11.08%	1.55%	894,848	0.0017
Flureka	305,200,000	77.77%	1.67%	45,401	0.0132
CNBC	31,200,000	165.13%	2.78%	12,172	0.0158
Chris Heria	4,220,000	75.11%	2.42%	1,227,584	0.0228
Chandoo	297,000	343.06%	1.68%	56,393	0.0404
Business Insider	68,700,000	111.01%	1.25%	589,631	0.0105
AehStudio7	5,700,000	58.33%	0.71%	4,857,387	0.0027

Managerial Interpretation

- Large subscriber counts do not always translate into high engagement. Some high-reach celebrity channels show lower Views per Subscriber, indicating passive audiences rather than active communities.
- Mid to large tier creators with strong Views per Subscriber outperform some top tier celebrities in engagement efficiency, making them potentially stronger partnership candidates.
- Average Views per Video varies significantly across creators, suggesting that consistency and content effectiveness differ even among large-scale channels.
- Music and entertainment channels dominate total reach, but engagement intensity is uneven, meaning not all large channels deliver the same campaign value.
- Creators with balanced performance across scale and engagement metrics represent the strongest partnership readiness profile.

Recommended Actions for Next Week

1. Prioritize outreach to creators who rank in the top tier for both Engagement Rate and Views per Subscriber rather than selecting solely based on subscriber count.
2. Segment creators into partnership tiers based on scale and efficiency to align them with appropriate brand campaign budgets.
3. Flag high-subscriber but low-efficiency channels for deeper qualitative review before recommending them for premium brand placements.

If a KPI Moves, What We Investigate

If the engagement rate declines, investigate recent content type, video duration, and recency of upload to determine whether audience interest or posting cadence has shifted. If Views per Subscriber drops, examine audience activation trends and review whether subscriber growth is outpacing engagement quality.

If Average Views per Video declines, analyze upload frequency, video format changes, and content mix adjustments. If Total Channel Subscribers increases but efficiency metrics decline, assess whether growth is organic and whether the audience remains engaged.

Digital Risks

Key Risks

- Vanity metric distortion. High subscriber counts may create the false impression of partnership value even if engagement efficiency is low.
- Engagement manipulation risk. Likes and comments can be artificially inflated through bots or coordinated activity, distorting true audience quality.
- Category and sampling bias. The dataset is heavily weighted toward music and celebrity channels, which may skew strategic decisions toward already dominant segments and overlook emerging creators.
- Heavy outliers can change the dependent fields and filters.

Mitigations

- Define clear KPI standards and benchmarks so decisions are not based on subscriber count alone but on engagement efficiency and content effectiveness.
- Implement validation checks by reviewing sudden spikes in likes, comments, or views before recommending high-budget partnerships.
- Establish ethical guardrails that require qualitative review alongside quantitative metrics to avoid reinforcing bias toward only large, established creators.

Team Roles + Contribution				
Frankie	Deshan	Tyler	Jorge	Jesús
Coordinated group communication and overall project flow throughout the assignment. Contributed to the Strategic Theme and Question, KPI development and definitions, Managerial Interpretation, Recommended Actions, and Digital Risks and Mitigations sections, ensuring alignment between strategy, metrics, and actionable brand partnership decisions.	Created the Tableau dashboard and led the visualization build. Sourced and prepared the dataset for analysis, ensuring the data was structured and ready for KPI calculation and dashboard integration.	Drafted and finalized the AI Use Statement section, clearly outlining how generative AI tools were used in accordance with course guidelines and ensuring transparency in methodology and compliance with academic integrity requirements.	Completed "Data and Metrics" section. Explained how the data source was learned based on key factors, i.e., views, subscribers, engagement, etc. Emphasized the limits of the dataset, such as potential metric inflation, outdated performance, and engagement risks.	Assisted in different sections as: - "Strategic Theme and Questions", - "Key fields used." - "KPI definitions" Assisted in completing the tables previously mentioned. Assisted in "Known limitations."
Group Collaborative Contributions				
All group members collaborated to develop the Business Scenario, Primary User, and Key Weekly Decisions. We worked together to align the strategic framing and ensure the dashboard supports realistic brand partnership and talent selection decisions.				

AI Use Statement

Generative AI tools were used to support ideation, structure refinement, and clarity of written explanations. Specifically, AI was used to help draft the business scenario, refine the strategic question, define KPIs with clear formulas, and organize the managerial interpretation and risk sections. AI was not used to generate or manipulate the dataset, fabricate results, or create the final Tableau visualizations.

Representative prompts included: “Refine this strategic question for clarity,” and “Suggest KPI definitions using these fields.” The outputs were reviewed, edited, and adapted to align with the dataset, assignment requirements, and course guidelines before inclusion in the final submission.